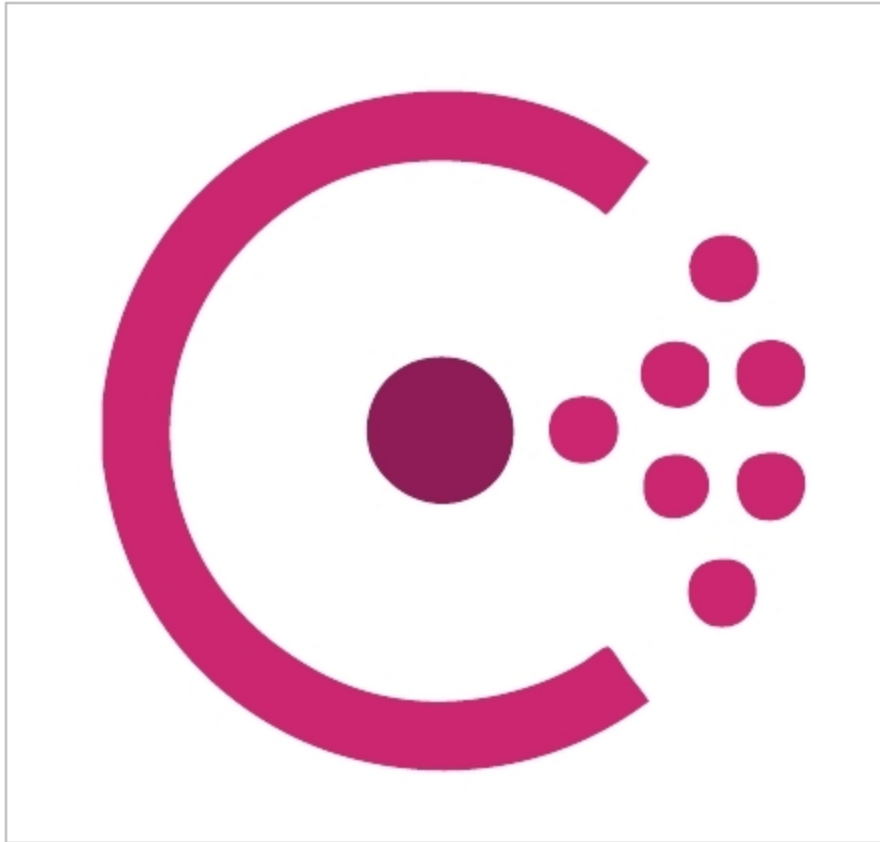


DevOps Series

Ansible Deployment of Consul

Consul supports multiple data centres, can be used as a key/value store, and can monitor cluster health. In this 17th article in the DevOps series, we discuss the Ansible deployment of Consul.



Consul is a tool that has been written by HashiCorp, and can be used for creating health checks for services and systems. It provides a simple HTTP API for using it as a key/value store. Consul is distributed, highly available, and is data centre aware. The recommended number of Consul nodes is three or five in a cluster, in order to handle failures. Every data centre can contain a Consul cluster. Consul is released under the Mozilla Public License 2.0.

Setting up Consul

We are going to set up and configure a Consul cluster with three Debian 9 (x86_64) guest virtual machines (VMs) using KVM/QEMU.

The host system is a Parabola GNU/Linux-libre x86_64 system and Ansible is installed using the distribution package manager. The version of Ansible used is 2.5.3 as indicated below:

```
$ ansible --version
ansible 2.5.3
  config file = /etc/ansible/ansible.cfg
  configured module search path = ['/home/guest/.ansible/
plugins/modules', '/usr/share/ansible/plugins/modules']
```

```
ansible python module location = /usr/lib/python3.6/site-
packages/ansible
executable location = /usr/bin/ansible
python version = 3.6.5 (default, May 11 2018, 04:00:52)
[GCC 8.1.0]
```

On the host system, we will create a project directory structure to store the Ansible playbooks, inventory and configuration files, as shown below:

```
ansible/inventory/kvm/
  /playbooks/configuration/
  /files/
```

The *inventory/kvm/inventory* file contains the following:

```
[consul1]
host1 ansible_host=192.168.122.140 ansible_connection=ssh
ansible_user=debian ansible_password=password

[consul2]
host2 ansible_host=192.168.122.208 ansible_connection=ssh
ansible_user=debian ansible_password=password

[consul3]
host3 ansible_host=192.168.122.59 ansible_connection=ssh
ansible_user=debian ansible_password=password

[bootstrap:children]
consul1

[server:children]
consul2
consul3
[all:children]
bootstrap
server
```

We have three Consul guest VMs running that are labelled as 'consul1', 'consul2' and 'consul3'. consul1 is defined as the bootstrap node. The other members (consul2 and consul3) of the Consul cluster belong to the server group. The 'all' group consists of all the three nodes.